Advanced Causal Inference

Lecture 05: Matching Estimators

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This lecture note is partially based on the lecture notes/textbook of Hyunseung Kang, Ying Zhou

Outline

1. Matched-Pairs Experiment
   - 1.1 Fisherian Inference
   - 1.2 Neymanian Inference
   - 1.3 Covariate adjustment
2. Matching in Observational Studies
3. Sensitivity Analysis
4. Assignment
5. Theoretical Aspects of Matching Estimators

Matched-Pairs Experiment (MPE)

- a special case of the stratified randomized experiment (SRE) with only one treated and one control unit per stratum (pair)
- strata are called pairs
- special inference and estimation strategies

Design of MPE

- Consider $2N$ units
- Pairing based on covariates:
  - Scalar covariate: Order and form adjacent pairs
  - Multiple covariates: Define pairwise distances and use matching algorithms
- Matching algorithms:
  - Greedy algorithm: pairs the two units with the smallest distance, drops them from the pool of units, pairs the two remaining units with the smallest distance, etc.
  - Optimal nonbipartite matching: divides the $2N$ units into N pairs of two units to minimize the sum of the within-pair distances

Notations

- Let (i, j) index the unit j in pair i, where $i = 1, \ldots, N$ and $j = 1, 2$
- Unit (i, j) has potential outcomes $Y_{ij}(1)$ and $Y_{ij}(0)$ under the treatment and control
- Within each pair, we randomly assign one unit to receive the treatment and the other to receive the control

$$A_i = \begin{cases} 1, & \text{if the first unit receives the treatment} \\ 0, & \text{if the second unit receives the treatment} \end{cases}$$

Treatment Assignment in MPE

Definition

Matched-Pairs Experiment (MPE):

$$(A_i)_{i=1}^N \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(1/2)$$

$$Y_{i1} = A_i Y_{i1}(1) + (1 - A_i) Y_{i1}(0)$$

$$Y_{i2} = A_i Y_{i2}(0) + (1 - A_i) Y_{i2}(1)$$

- Observed data: $(A_i, Y_{i1}, Y_{i2})_{i=1}^N$

Fisher Randomization Test (FRT)

- **Sharp null hypothesis:**

$$H_{0F} : Y_{ij}(1) = Y_{ij}(0), \text{ for all } i = 1, \ldots, N \text{ and } j = 1, 2$$

- Simulate the distribution of $(A_1, \ldots, A_N)$

- Compute test statistics based on within-pair differences:

$$\begin{aligned} \widehat{\tau}_i &= \text{outcome under treatment} - \text{outcome under control (within pair } i) \\ &= (2A_i - 1)(Y_{i1} - Y_{i2}) \\ &= S_i(Y_{i1} - Y_{i2}), \quad S_i = 2A_i - 1. \end{aligned}$$

- $S_i = 2A_i - 1$ are i.i.d. random signs with mean 0 and variance 1, for $i = 1, \ldots, N$.

Test Statistic

Estimate:

$$\widehat{\tau} = \frac{1}{N} \sum_{i=1}^{N} \widehat{\tau}_i$$

Variance under H_{0F}:

$$\mathrm{Var}(\widehat{\tau}) = \frac{1}{N^2} \sum_{i=1}^{N} \widehat{\tau}_i^2$$

Test statistic: Based on the CLT for the sum of independent random variables, we have the Normal approximation:

$$\frac{\widehat{\tau}}{\sqrt{N^{-2} \sum_{i=1}^{N} \widehat{\tau}_i^2}} \xrightarrow{D} N(0,1)$$

Wilcoxon Signed-Rank Test

- Uses the ranks $(R_1, \ldots, R_N)$ of $(|\hat{\tau}_1|, \ldots, |\hat{\tau}_N|)$

- Test statistic:

$$W = \sum_{i=1}^{N} \mathbb{1}(\hat{\tau}_i > 0) R_i.$$

Expectation and variance under H_0:

$$\mathrm{E}(W) = \frac{1}{2}\sum_{i=1}^{N} R_i = \frac{1}{2}\sum_{i=1}^{N} i = \frac{N(N+1)}{4}\ , \qquad \mathrm{Var}(W) = \frac{1}{4}\sum_{i=1}^{N} R_i^2 = \frac{1}{4}\sum_{i=1}^{N} i^2 = \frac{N(N+1)(2N+1)}{24}$$

- Approximate normality via CLT:

$$\frac{W - N(N+1)/4}{\sqrt{N(N+1)(2N+1)/24}} \to N(0,1)$$

- Implemented in R using `wilcox.test(..., paired=TRUE)`

McNemar's Test for Binary Outcomes

- Used when the outcome is binary (e.g., success/failure)
- Data is summarized in a 2x2 table:

	Control Outcome 1	Control Outcome 0
Treated Outcome 1	m_{11}	m_{10}
Treated Outcome 0	m_{01}	m_{00}

- Under $\mathcal{H}_F$, the numbers of concordant pairs m_{11} and m_{00} are fixed; $m_{10} + m_{01}$ is also fixed; the only random component is m_{10}
- Exact test based on Binomial distribution:

$$m_{10} \sim \text{Binomial}(m_{10} + m_{01}, 1/2)$$

- Asymptotic test (Normal approximation):

$$\frac{m_{10} - m_{01}}{\sqrt{m_{10} + m_{01}}} \to \mathcal{N}(0, 1)$$

- Implemented in R: `mcnemar.test`

Neymanian Inference

- The average causal effect within pair i is:

$$\tau_i = \frac{1}{2}\left\{Y_{i1}(1) + Y_{i2}(1) - Y_{i1}(0) - Y_{i2}(0)\right\}.$$

- The overall average causal effect is:

$$\tau = \frac{1}{N}\sum_{i=1}^{N}\tau_i = \frac{1}{2N}\sum_{i=1}^{N}\sum_{j=1}^{2}\left\{Y_{ij}(1) - Y_{ij}(0)\right\}.$$

- The estimators:

$$\widehat{\tau}_i = (2A_i - 1)(Y_{i1} - Y_{i2}) = A_i\{Y_{i1}(1) - Y_{i2}(0)\} + (1 - A_i)\{Y_{i2}(1) - Y_{i1}(0)\}$$

$$\widehat{\tau} = \frac{1}{N}\sum_{i=1}^{N}\widehat{\tau}_i$$

- Question: Are $\widehat{\tau}_i$ and $\widehat{\tau}$ unbiased estimators of τ_i and τ?

Variance of $\widehat{\tau}$

- Note that the MPE is just a special case of the SRE

- Within pair i, the difference-in-means $\widehat{\tau}_i$ is unbiased for τ_i with variance

$$\mathrm{Var}(\widehat{\tau}_i) = S_i^2(1) + S_i^2(0) - \frac{S_i^2(\tau)}{2}\ , \qquad S_i^2(a) = \sum_{j=1}^{2} \{Y_{ij}(a) - \bar{Y}_i(a)\}^2 \quad S_i^2(\tau) = \sum_{j=1}^{2} (\tau_{ij} - \tau_i)^2$$

- The estimator $\widehat{\tau} = \frac{1}{N}\sum_{i=1}^{N} \widehat{\tau}_i$ has variance $\mathrm{Var}(\widehat{\tau}) = \frac{1}{N^2}\sum_{i=1}^{N} \mathrm{Var}(\widehat{\tau}_i)$

- Within-pair sample variances are not defined due to one treated and one control unit per pair
  $\Rightarrow$ cannot estimate $\mathrm{Var}(\widehat{\tau}_i)$!

- Change the perspective to the classic i.i.d. sampling
  - If the $\widehat{\tau}_i$'s are IID with mean μ and σ^2, then it is well-known that $\mathrm{E}(\widehat{V}) = \mathrm{Var}(\widehat{\tau}) = \sigma^2/N$ where

$$\widehat{V} = \frac{1}{N(N-1)} \sum_{i=1}^{N} (\widehat{\tau}_i - \widehat{\tau})^2$$

- This is not true under the finite-population framework

Conservativeness of $\widehat{V}$

Theorem

Under the MPE, $\widehat{V}$ is a conservative estimator of $\mathrm{Var}(\hat{\tau})$:

$$\mathrm{E}(\widehat{V}) - \mathrm{Var}(\hat{\tau}) = \frac{1}{n(n-1)} \sum_{i=1}^{n} (\tau_i - \tau)^2 \geqslant 0.$$

If the τ_i's are constant across pairs, then $\mathrm{E}(\widehat{V}) = \mathrm{Var}(\hat{\tau})$.

- Using the identity: $\sum_{i=1}^{N}(a_i - \bar{a})^2 = \sum_{i=1}^{N} a_i^2 - N\bar{a}^2$
- Applying this to $\widehat{V}$:

$$\begin{aligned} N(N-1)\mathrm{E}(\widehat{V}) &= \mathrm{E}\left\{ \sum_{i=1}^{N} (\hat{\tau}_i - \hat{\tau})^2 \right\} = E\left(\sum_{i=1}^{N} \hat{\tau}_i^2 - N\hat{\tau}^2 \right) \\ &= \sum_{i=1}^{N} \{\mathrm{Var}(\hat{\tau}_i) + \tau_i^2\} - N\{\mathrm{Var}(\hat{\tau}) + \tau^2\} = \sum_{i=1}^{N} \mathrm{Var}(\hat{\tau}_i) - N\mathrm{Var}(\hat{\tau}) + \sum_{i=1}^{N} \tau_i^2 - N\tau^2 \\ &= N^2\mathrm{Var}(\hat{\tau}) - N\mathrm{Var}(\hat{\tau}) + \sum_{i=1}^{N} (\tau_i - \tau)^2 \end{aligned}$$

- Simplifying gives: $E(\widehat{V}) = \mathrm{Var}(\hat{\tau}) + \dfrac{1}{n(n-1)} \sum_{i=1}^{n} (\tau_i - \tau)^2 \geqslant \mathrm{Var}(\hat{\tau})$

Large-Sample Approximation

- Asymptotically, $\widehat{\tau}$ follows a normal distribution:

$$\frac{\widehat{\tau} - \tau}{\sqrt{\mathrm{Var}(\widehat{\tau})}} \xrightarrow{D} N(0, 1)$$

- Due to the over-estimation of the variance, the Wald-type confidence interval

$$\widehat{\tau} \pm z_{1-\alpha/2}\sqrt{\widehat{V}}$$

covers τ with probability at least $1 - \alpha$ (but is conservative in general)

- $\widehat{\tau}$ and $\widehat{V}$ are identical to the coefficient and variance estimator of the intercept from the OLS fit of the vector $(\widehat{\tau}_1, \ldots, \widehat{\tau}_N)^\intercal$ on the intercept only

Covariate Adjustment

- Matching on covariates in the design stage may not be perfect ($X_{i1} \neq X_{i2}$).
- Additional covariates beyond those used in matching may be available.
- Adjusting for covariates can improve *estimation efficiency*
- Two main strategies for covariate adjustment in MPE:
  - Fisher Randomization Test
  - Regression Adjustment

Covariate-Adjusted Fisher Randomization Test

- Construct test statistics based on residuals from outcome regression on covariates
- Residuals are fixed under the sharp null hypothesis
- Approach:

1. Fit an *OLS model* of all observed outcomes on covariates.
2. Compute residuals $\widehat{\varepsilon}_{ij}$
3. Conduct randomization inference using these residuals

Regression Adjustment for Causal Estimation

- Define the within-pair covariate difference:

$$\widehat{\tau}_{X,i} = (2A_i - 1)(X_{i1} - X_{i2}), \quad \widehat{\tau}_X = \frac{1}{N}\sum_{i=1}^{N} \widehat{\tau}_{X,i}$$

- Properties: $\mathrm{E}(\widehat{\tau}_{X,i}) = 0$, $\mathrm{E}(\widehat{\tau}_X) = 0$, $\mathrm{Cov}(\widehat{\tau}_X) = \frac{1}{N^2}\sum_{i=1}^{N} \widehat{\tau}_{X,i}\widehat{\tau}_{X,i}^{\intercal}$

- Imbalance in covariate means can impact efficiency

- Regression adjustment improves efficiency by reducing imbalance

Optimal Covariate Adjustment Strategy

- Consider a family of estimators indexed by γ: $\hat{\tau}(\gamma) = \hat{\tau} - \gamma^\intercal \hat{\tau}_X$
- $E\{\hat{\tau}(\gamma)\} = \tau$ for any γ
- Its variance is a quadratic function of γ:

$$\mathrm{Var}\{\hat{\tau}(\gamma)\} = \mathrm{Var}(\hat{\tau} - \gamma^\intercal \hat{\tau}_X) = \mathrm{Var}(\hat{\tau}) + \gamma^\intercal \mathrm{Cov}(\hat{\tau}_X)\gamma - 2\gamma^\intercal \mathrm{Cov}(\hat{\tau}_X, \hat{\tau})$$

- Minimize the variance: $\tilde{\gamma} = \mathrm{Cov}(\hat{\tau}_X)^{-1}\mathrm{Cov}(\hat{\tau}_X, \hat{\tau}) \quad \Rightarrow \quad$ an estimator of $\tilde{\gamma} = \hat{\gamma} = \widehat{\mathrm{Cov}}(\hat{\tau}_X)^{-1}\widehat{\mathrm{Cov}}(\hat{\tau}_X, \hat{\tau})$
  - FACT: $\mathrm{Cov}(\hat{\tau}_X) = \frac{1}{N^2}\sum_{i=1}^N |\hat{\tau}_{X,i}||\hat{\tau}_{X,i}|^\intercal$ where $|\cdot|$ denotes component-wise absolute value of a vector
  - $\mathrm{Cov}(\hat{\tau}_X)$ is fixed and known from observed data
  - However, $\mathrm{Cov}(\hat{\tau}_X, \hat{\tau})$ depends on unknown potential outcomes
  - An unbiased estimator for $\mathrm{Cov}(\hat{\tau}_X, \hat{\tau})$ is $\widehat{\mathrm{Cov}}(\hat{\tau}_X, \hat{\tau}) = \frac{1}{N(N-1)}\sum_{i=1}^N (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau})$

Covariance Estimation and Optimal Adjustment

- Estimate optimal coefficient $\tilde{\gamma}$:

$$\hat{\gamma} = \left(\frac{1}{N^2} \sum_{i=1}^{N} \hat{\tau}_{X,i} \hat{\tau}_{X,i}^{\intercal} \right)^{-1} \left\{ \frac{1}{N(N-1)} \sum_{i=1}^{N} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau}) \right\} \approx \left(\sum_{i=1}^{N} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_{X,i} - \hat{\tau}_X)^{\intercal} \right)^{-1} \sum_{i=1}^{N} (\hat{\tau}_{X,i} - \hat{\tau}_X)(\hat{\tau}_i - \hat{\tau})$$

  $\Rightarrow$ approximately the coefficient of the $\hat{\tau}_{X,i}$ in the OLS fit of the $\hat{\tau}_i$'s on the $\hat{\tau}_{X,i}$'s with an intercept

- Final estimator: $\hat{\tau}_{\text{adj}} = \hat{\tau} - \hat{\gamma}^{\intercal} \hat{\tau}_X \simeq$ intercept of $\texttt{lm}(\hat{\tau}_i \sim 1 + \hat{\tau}_{X,i})$

- Conservative variance estimator:

$$\widehat{V}_{\text{adj}} = \widehat{V} + \hat{\gamma}^{\intercal} \widehat{\text{Cov}}(\hat{\tau}_X) \hat{\gamma} - 2\hat{\gamma}^{\intercal} \widehat{\text{Cov}}(\hat{\tau}_X, \hat{\tau}) = \widehat{V} - \widehat{\text{Cov}}(\hat{\tau}_X, \hat{\tau})^{\intercal} \text{Cov}(\hat{\tau}_X)^{-1} \widehat{\text{Cov}}(\hat{\tau}_X, \hat{\tau})$$

- Improves precision over unadjusted variance estimator

- Variance estimator $\widehat{V}_{\text{adj}}$ is the corresponding OLS variance estimate

- Fogarty (2018): not requiring the EHW correction; no need to center the covariates
  - Because we reduce the data from the MPE to the within-pair differences, it is unnecessary to center the covariates, which is different from the implementation of Lin (2013)'s estimator under the CRE.

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1 Matched-Pairs Experiment
1.1 Fisherian Inference
1.2 Neymanian Inference
1.3 Covariate adjustment

2 Matching in Observational Studies

3 Sensitivity Analysis

4 Assignment

5 Theoretical Aspects of Matching Estimators

Matching in Observational Studies

- **Goal:** To estimate causal effects by creating comparable treatment and control groups in observational studies
- Unlike randomized experiments, treatment assignment in observational studies is not random
- Matching attempts to create pairs (or groups) of treated and control units with similar characteristics
- This reduces confounding and makes causal inference more reliable
- See Stuart (2010) for a nice summary
- We will use `MatchIt::lalonde` data as an example (LaLonde, 1986; Dehejia and Wahba, 1999)

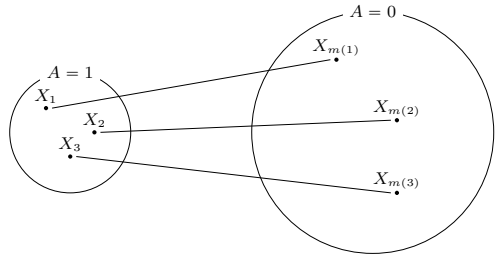
Advantages and Limitations

- **Advantages**:
  - Simple and intuitive: No advanced math skills are needed to grasp the logic
  - Non-parametric: Requires minimal assumptions on the observed data distribution
  - Blind to outcomes: Prevents p-hacking
  - Diagnostics are easy: Overlap can be directly assessed

- **Limitations**:
  - Theoretical complexity: Establishing theory is challenging and still under-developed
  - Requires practice: Matching is not like regression; you need practice and experience to get good matches
  - Estimand: Matching usually estimates the ATT/ATC, not the ATE,
    since it applies only to the matched sample unless explicitly designed otherwise

Pair Matching

- **Assignment Problem**
  - Units must be paired: (Bob, Scott), (Kathleen, Jennifer)
  - Each pair: one treated unit ($A = 1$) and one control ($A = 0$)



- **Matching as Optimization**
  - Minimum cost flow problem
  - Solved using the Hungarian algorithm (aka Kuhn–Munkres algorithm)
  - Complexity: $O(\max(n_T, n_C)^3)$

- **Causal Inference Logic**
  - Compare outcomes between paired treated and control units
  - Differences $\Rightarrow$ causal effect (assuming no unmeasured confounding)

Types of Pair Matching

Type	How It Works
Exact Matching	Matches treated and control units only if their covariates are identical
Nearest Neighbor Matching	Matches each treated unit to the control unit with the smallest distance in covariate space (e.g., Mahalanobis distance, propensity score distance)
Caliper/Radius Matching	Matches treated and control units if their propensity scores differ by less than a specified threshold

Exact Matching

- **Definition:** Each treated unit is matched with a control unit such that covariates are identical
- Given treated units $i = 1, \ldots, N_1$ and a larger pool of control units $N_0 \gg N_1$, we find a match $m(i)$ for each i such that:

$$X_i = X_{m(i)}$$

- In this ideal case, matched units share the same propensity score:

$$\Pr(A_i = 1|X_i) = e(X_i) = e(X_{m(i)}) = \Pr(A_{m(i)} = 1|X_{m(i)})$$

- This ensures balance between treatment and control groups
  - Recall the balancing property of the propensity score: $\{Y(a), X\} \perp\!\!\!\perp A \,|\, e(X)$ under cond. ign.

Propensity Score and Treatment Assignment

- **Key Probability Argument:**

- Conditional on the event that one unit receives treatment ($A_i = 1$) and the other does not ($A_{m(i)} = 0$), symmetry implies:

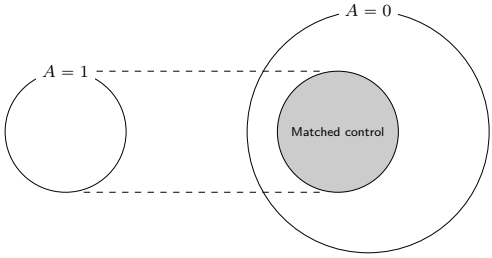
$$\Pr(A_i = 1, A_{m(i)} = 0 \mid A_i + A_{m(i)} = 1, X_i, X_{m(i)}) = \frac{1}{2}$$

- This follows from the detailed probability calculation:

$$\begin{aligned}
&\Pr(A_i = 1, A_{m(i)} = 0 \mid A_i + A_{m(i)} = 1, X_i, X_{m(i)}) \\
&= \frac{\Pr(A_i = 1, A_{m(i)} = 0 \mid X_i, X_{m(i)})}{\Pr(A_i = 1, A_{m(i)} = 0 \mid X_i, X_{m(i)}) + \Pr(A_i = 0, A_{m(i)} = 1 \mid X_i, X_{m(i)})} \\
&= \frac{e(X_i)\{1 - e(X_{m(i)})\}}{e(X_i)\{1 - e(X_{m(i)})\} + \{1 - e(X_i)\}e(X_{m(i)})} \\
&= \frac{1}{2}
\end{aligned}$$

Matched Pair Experiment (MPE) Analysis

- Exact matching allows observational studies to be analyzed like a Matched Pair Experiment (MPE).
- We can use:
  - Fisher Randomization Test (FRT) for hypothesis testing.
  - Neymanian Approach for estimating causal effects.
- This provides valid inference for the causal effect on the treated units
  - Exact matching estimates the counterfactual outcomes for treated units using matched controls, so it gives the ATT



Exact Matching

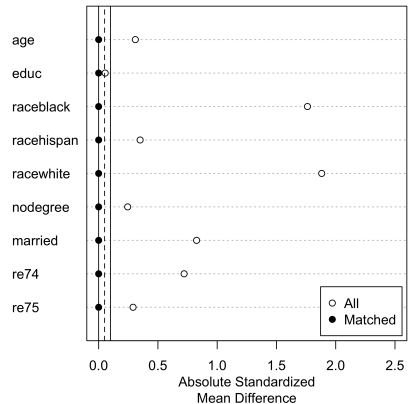
- ATT estimate = -1106.3

```
library(MatchIt)
data("lalonde")

LL.Exact.match <-
  matchit(treat ~ age + educ + race + nodegree + married + re74 + re75,
          data = lalonde, method = "exact", estimand="ATT")
plot(summary(LL.Exact.match),xlim=c(0,2.5))
LL.Data.Exact.match <- match_data(LL.Exact.match)
LM.Exact <-  lm(re78~treat,data=LL.Data.Exact.match,weights=weights)

###################################

            Estimate Std. Error t value Pr(>|t|)
(Intercept)     6092       1888   3.227  0.00373 **
treat          -1106       2618  -0.423  0.67652
```



Exact Matching

- One can consider a cluster-robust SE:

```
library(sandwich; library(lmtest)

coeftest(LM.Exact, vcov = vcovHC(LM.Exact, type = "HC1"))

##################################

             Estimate Std. Error t value Pr(>|t|)
(Intercept)   6091.7     2064.8  2.9503 0.007179 **
treat        -1106.3     2735.5 -0.4044 0.689631
```

- `t.test(re78 treat,data=LL.Data.Exact.match)` does not give the same result

```
 mean in group 0 mean in group 1
       6590.784        4985.390
```

- So, recommend using `lm(Y∼A,weights)`

Limitations of Exact Matching

- **Reality:** In most observational studies, exact matching ($X_i = X_{m(i)}$) is not possible

```
sum(LL.Data.Exact.match$treat)
[1] 13

sum(lalonde$treat)
[1] 185
```

- Rosenbaum (2002) supports the matched analysis strategy

- However, recent work by Guo and Rothenhäusler (2022) warns that inexact matching may lead to biased inference in FRT

- This highlights the need for sensitivity analysis and robustness checks

Nearest Neighbor Matching

- Instead of exact matching, we aim for approximate matches where $X_i \approx X_{m(i)}$
- Approximate matching minimizes the difference between matched pairs based on a distance metric
- **How do we find the closest match?**
  - For each treated unit i, find a control unit $m(i)$ that minimizes:

$$\text{nearest neighbor of } i = m(i) = \underset{k:A_k=0}{\arg\min}\, d(X_i, X_k)\ , \quad d(\cdot,\cdot) = \text{distance metric}$$

- Common distance metrics include:
  - Euclidean distance: $d(X_i, X_k) = (X_i - X_k)^\intercal (X_i - X_k)$
  - Mahalanobis distance: $d(X_i, X_k) = (X_i - X_k)^\intercal S_X^{-1} (X_i - X_k)$
    S_X = sample covariance matrix, estimated from the population or the control group
  - Propensity score distance: $d(X_i, X_k) = |\hat{e}(X_i) - \hat{e}(X_k)|$ or $d(X_i, X_k) = |\text{logit}\{\hat{e}(X_i)\} - \text{logit}\{\hat{e}(X_k)\}|$
    $\hat{e}$ = estimated propensity score; say `glm(A~X, family="binomial")`
- Rosenbaum and Rubin (1985): recommended the logit scale rather because $\hat{e}$ often cluster near 0 and 1 while the logit propensity scores remain more dispersed

Why Propensity Score-based Matching?

- If $X_i \overset{iid}{\sim} N(0, I_p)$, then $Q = (X_i - X_k)^\intercal (X_i - X_k) \sim 2\chi^2_p$
- $\mathrm{E}(Q) = 2p$, $\mathrm{Var}(Q) = 8p$
- As p increases, differences between units become large, finding good matches becomes difficult
- With large p, imperfect matching often introduces large bias in causal effect estimation
- If p is large, we must do some dimension reduction before matching, say propensity score-based matching

Nearest Neighbor Matching

- **R-Implementation:**

```
library(MatchIt)
data("lalonde")

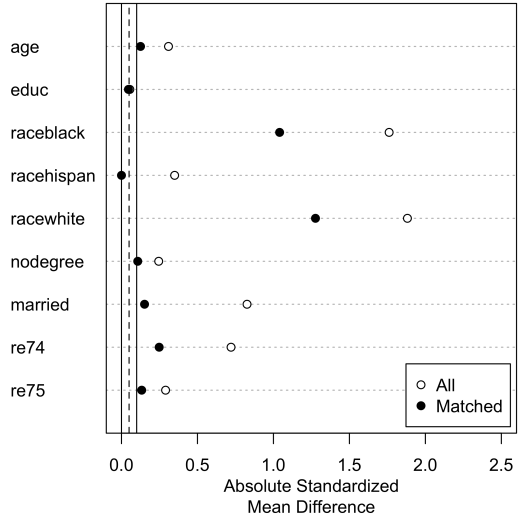
LL.M.match <- matchit(treat ~ age + educ + race + nodegree + married + re74 + re75,
                      data = lalonde,
                      method = "nearest",
                      distance = "mahalanobis",
                      estimand = "ATT")

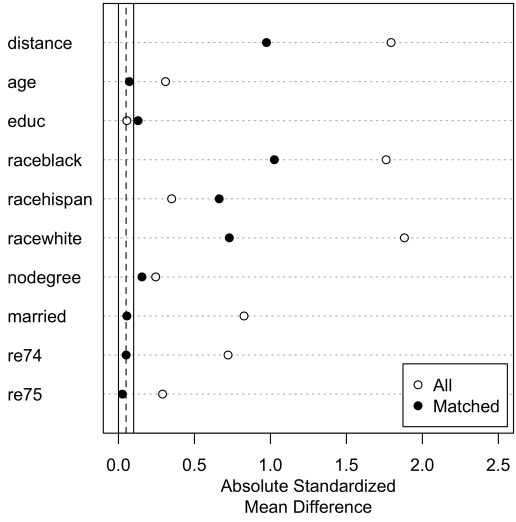
LL.PS.match <- matchit(treat ~ age + educ + race + nodegree + married + re74 + re75,
                       data = lalonde,
                       method = "nearest",
                       distance = "logit",
                       estimand = "ATT")

plot(summary(LL.M.match),xlim=c(0,2.5))

plot(summary(LL.PS.match),xlim=c(0,2.5))
```

Nearest Neighbor Matching





Mahalanobis & Propensity score

Nearest Neighbor Matching

- ATT estimate = 538.563

```
LL.Data.M.match <- match_data(LL.M.match)
summary( lm(re78~treat,data=LL.Data.M.match,weights=weights) )

###################################

            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5810.6      528.2  11.000   <2e-16 ***
treat          538.6      747.1   0.721    0.471
```

- ATT estimate = 894.368

```
LL.Data.PS.match <- match_data(LL.PS.match)
summary( lm(re78~treat,data=LL.Data.PS.match,weights=weights) )

###################################

            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5454.8      516.4  10.563   <2e-16 ***
treat          894.4      730.3   1.225    0.221
```

Combining Distance Measures

- The distance measures described above can also be combined. For example:

- Exact matching on key covariates (e.g., race) + propensity score matching within those groups

```
matchit(treat ~ age + educ + race + married + nodegree + re74 + re75, data = lalonde,
        method = "nearest", distance = "logit", exact = ~ race)
```

- When exact matching on even a few variables is not possible,

  "fine balance" can be considered (e.g., the same proportion of White in the matched treated and control groups)

```
library(optmatch)
ps_dist <- match_on(treat ~ age + educ + re74 + re75, data = lalonde)
matched <- pairmatch(ps_dist, fine = lalonde$race)
```

Caliper Matching

- **Caliper Matching** (Rubin and Thomas, 2000):
  Mahalanobis matching of the key variables Z within propensity score calipers:

$$d(X_i, X_k) = \begin{cases} (Z_i - Z_k)^\intercal S_Z^{-1} (Z_i - Z_k) & \text{if } |\text{logit}\{\hat{e}(X_i)\} - \text{logit}\{\hat{e}(X_k)\}| \leqslant c \\ \infty & \text{if } |\text{logit}\{\hat{e}(X_i)\} - \text{logit}\{\hat{e}(X_k)\}| > c \end{cases}$$

  - This will yield matches that are relatively well matched on the propensity score and particularly well matched on Z

```
matchit(treat ~ age + educ + race + married + nodegree + re74 + re75, data = lalonde,
        method = "nearest", distance = "logit",
        caliper = 0.2, std.caliper = TRUE,
        mahvars = ~ age + educ)
```

  - $c = 0.2 \times \text{sd}(\text{logit}(\hat{e}))$ and $Z = (\text{age}, \text{educ})$

Limitations of 1-to-1 Nearest Neighbor Matching/Caliper Matching

- Not Globally Optimal & Order Sensitivity
  - Nearest neighbor matching is a greedy algorithm, so it may result in poor quality matches overall
  - The order of treated units affects which controls are matched
  - The first few matches might be good matches, and the rest poor matches

- Sample Size Reduction
  - Implementing discarding unmatched units can remove treated units from analysis
  - While this improves match quality, it reduces effective sample size
  - Reduced sample size can decrease statistical power and generalizability

- Sensitivity to Outliers/Caliper Choice
  - Extreme values in covariates or propensity scores, and a specific choice of caliper can distort matches
  - This can increase bias, particularly in datasets with heavy tails or unbalanced distributions

Optimal Pair Matching

- **Usual (Greedy) Pair Matching**: `matchit(..., method="nearest")`

  Matches each treated unit sequentially to the nearest available control based on distance

- **Optimal Pair Matching**: `matchit(..., method="optimal")`

  Selects matched pairs to minimize the sum of absolute pairwise distances in the sample

  - Matching order does not affect results, unlike greedy nearest neighbor matching
  - Reduces the risk of extreme within-pair distances
  - Performs similarly to nearest neighbor matching in selecting subsets, but better at minimizing distances within pairs

- Greedy matching may be sufficient for creating balanced groups
- Optimal matching is preferable when the goal is well-matched pairs

Matching with and without Replacement

- **Matching with Replacement:** `matchit(..., replace=TRUE)`
  - A control unit can be matched multiple times
  - Computationally easier
  - Produces higher-quality matches
  - Introduces dependence in the analysis

- **Matching without Replacement:** `matchit(..., replace=FALSE)`
  - Each control unit is used only once
  - Leads to independent matched pairs
  - Requires solving a complex discrete optimization problem

- If the control pool is large, the choice between these methods has minimal impact

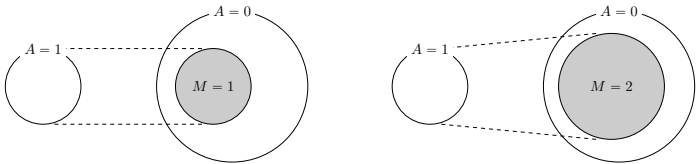
One-to-Many Matching

- **One-to-One Matching**: Each treated unit is matched to exactly one control unit

- When the control pool is large, allowing more matches for some treated units
  - Increases statistical power by using more control information
  - Requires careful handling of correlated data, since matched controls are likely to be dependent due to similar X

- Smith (1997): **Ratio Matching** or **One-to-M Matching**: `matchit(..., ratio = M)`
  - The number of matched controls per treated unit (M) is fixed across all treated units

- Ming and Rosenbaum (2001): **Variable-Ratio Matching** or **One-to-M_i Matching**:
  - The number of matched controls per treated unit (M_i) varies across treated units



Full Matching: Many-to-Many Matching

- **Subclassification** groups individuals with similar characteristics, often using quintiles of the propensity score
- **Full matching** (Rosenbaum, 1991; Hansen, 2004; Stuart and Green, 2008): `matchit(..., method="full")`
  - An advanced form of subclassification that automatically selects the number of subclasses
  - Optimal for minimizing the average distance between treated and control individuals within matched sets
  - Matched sets containing at least one treated and one control individual: (2T:1C), (1T:1C), (1T:2C), ...
  - Can estimate either the ATE or ATT: `estimand=''ATE''` or `estimand=''ATT''`
  - We often focus on the ATE for full matching; in this case, need to use `estimand=''ATE''` because the default option is `estimand=''ATT''`

Name	Propensity Score	Quintile Group	Treated	Matched Set
Alice	0.12	1	1	A (2T:1C)
Fiona	0.29	2	1	A (2T:1C)
Ian	0.05	1	0	A (2T:1C)
Charlie	0.47	3	1	B (1T:2C)
Bob	0.35	2	0	B (1T:2C)
George	0.53	3	0	B (1T:2C)
Hannah	0.75	4	1	C (1T:1C)
Diana	0.68	4	0	C (1T:1C)
Ethan	0.91	5	1	D (1T:1C)
Julia	0.88	5	0	D (1T:1C)
		...		

Revisit: LaLonde

- ATT estimate = 1780.178

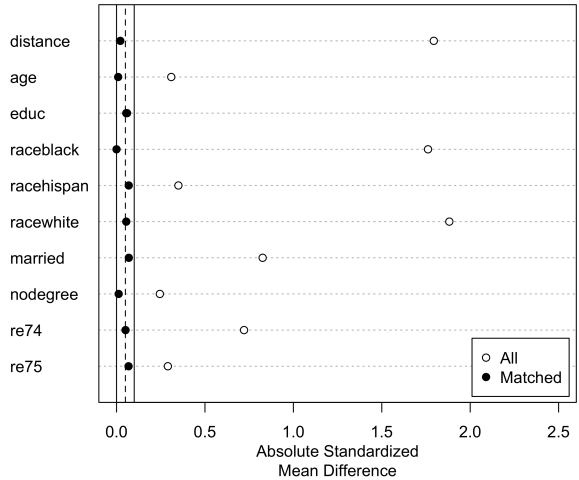
```
LL.F.match <-
  matchit(treat ~ age + educ + race + married +
            nodegree + re74 + re75,
          data = lalonde,
          method = "nearest", distance = "logit",
          caliper = 0.2, std.caliper = TRUE,
          mahvars = ~ age + educ + nodegree,
          replace=TRUE,
          estimand = "ATT")

plot(summary(LL.F.match),xlim=c(0,2.5))

summary( lm(re78~treat,data=LL.Data.F.match,weights=weights) )

###################################

             Estimate Std. Error t value Pr(>|t|)
(Intercept)    4533.4      806.8   5.619 4.91e-08 ***
treat          1816.1      966.4   1.879   0.0613 .
```



Revisit: LaLonde

- ATT estimate = 1780.178

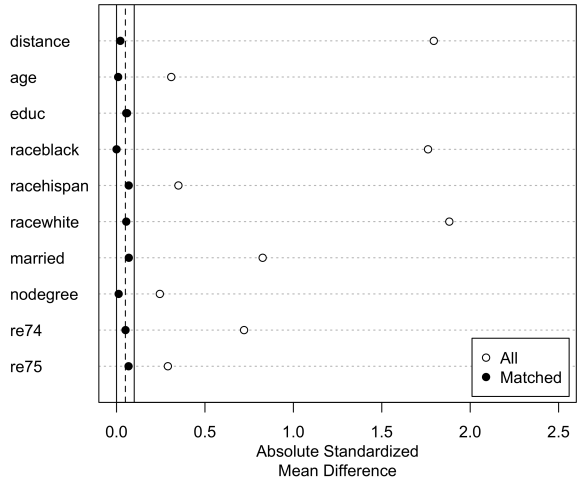
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Sensitivity Analysis for Matched Pairs with A Binary Outcome

- Rosenbaum (2002, Chapter 4.2)

- **Example:** Smoking and lung cancer.
  - Treatment A: 1 for smoking, 0 for not smoking.
  - Outcome Y: 1 for lung cancer, 0 for no lung cancer.

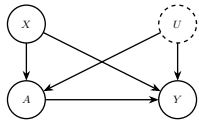
- **Setup:** We have I matched pairs, e.g., (Bob, Scott) is pair $S = 1$

Matched Set	Index of matched sets
Bob — Scott	$s = 1$
Jennifer — Kathleen	$s = 2$
...	...
...	$s = S$

- We assume conditional ignorability: $Y_i(a) \perp\!\!\!\perp A_i | X_i$, $\forall i$
  - Under this assumption, for any pair $s = 1, \ldots, S$, the probability of treatment assignment is equal
  - $p_s = \Pr(\text{person 1 gets treatment in pair } S) = 0.5$

Sensitivity Analysis for Matched Pairs with A Binary Outcome

- **Unmeasured Confounding:**
  - Suppose an unmeasured confounder U (e.g., a gene) makes Bob more likely to smoke than Scott



  - Then $p_s \neq 0.5$, and the assignment is no longer random given X_s
  - The probability p_s is now bounded by a sensitivity parameter $\Gamma \geqslant 1$:

$$\frac{1}{1+\Gamma} \leqslant p_s \leqslant \frac{\Gamma}{1+\Gamma}$$

  - This can be rewritten as a bound on the odds of treatment assignment:

$$\frac{1}{\Gamma} \leqslant \frac{p_s}{1-p_s} \leqslant \Gamma$$

Sensitivity Analysis for Matched Pairs with A Binary Outcome

- **Hypothesis Testing:**
  - We test the sharp null hypothesis: $H_0 : Y_i(1) = Y_i(0),\ \forall i$
  - The test statistic is the sum of the differences in outcomes: $T = \sum_{s=1}^{S} \delta_s$ where $\delta_s = Y_{\text{trt},s} - Y_{\text{ctrl},s}$

- **Null Distribution of T:**
  - Suppose in discordant pair s, one unit has outcome 1 and the other has outcome 0
  - If the unit with 1 is treated, then $Y_{\text{trt},s} - Y_{\text{ctrl},s} = 1 - 0 = +1$
  - If the unit with 0 is treated, then $Y_{\text{trt},s} - Y_{\text{ctrl},s} = 0 - 1 = -1$
  - So under the null, δ_s for the discordant pair s and T is given by

$$\delta_s = Q_s = \begin{cases} 1 & \text{w.p. } p_s \\ -1 & \text{w.p. } 1 - p_s \end{cases} \quad \Rightarrow \quad T = \sum_{s:\text{discordant}} Q_s$$

Sensitivity Analysis for Matched Pairs with A Binary Outcome

- We can write

$$T = \sum_{s:\text{discordant}} Q_s \quad \Rightarrow \quad \frac{T+d}{2} = \sum_{s:\text{discordant}} \frac{Q_s+1}{2} \quad \stackrel{D}{=} \sum_{s:\text{discordant}} b_s \ , \quad b_s \sim \mathsf{Ber}(p_s)$$

- Under $p_s = 0.5$, the one-sided p-value is given by

$$\text{p-value} = \Pr\left(\frac{T+d}{2} \geqslant \frac{T_{\text{obs}}+d}{2}\right) = \Pr\left\{\mathsf{Bin}\left(d, \frac{1}{2}\right) \geqslant \frac{T_{\text{obs}}+d}{2}\right\}$$

- Now, suppose that one unit in each pair is more likely to get treatment than the other due to U:

$$\frac{1}{1+\Gamma} \leqslant p_s \leqslant \frac{\Gamma}{1+\Gamma}$$

- This corresponds to the following worst-case distributions

$$\Pr\left\{\mathsf{Bin}\left(d, \frac{1}{1+\Gamma}\right) \geqslant \frac{T_{\text{obs}}+d}{2}\right\} \leqslant \Pr\left(\frac{T+d}{2} \geqslant \frac{T_{\text{obs}}+d}{2}\right) \leqslant \Pr\left\{\mathsf{Bin}\left(d, \frac{\Gamma}{1+\Gamma}\right) \geqslant \frac{T_{\text{obs}}+d}{2}\right\}$$

Example

- Suppose the study units are paired where each pair consists of one treated unit and a control unit and we are interested in testing the sharp null hypothesis of no treatment effect.

- There were $S = 36,975$ pairs
  - 122 pairs were discordant where one of the pair had lung cancer
    Among 122 pairs, 110 of these pairs had the smokers developing lung cancer
    Among 122 pairs, 12 of these pairs had the non-smoker developing lung cancer
  - 36,853 pairs had same outcomes between the pairs

$Y_{\text{smoking}} - Y_{\text{no smoke}}$	Frequency
1	110
-1	12
0	36975

Example

- The one-sided p-value is given by

$$\text{p-value} = \Pr\left(\frac{T+122}{2} \geqslant \frac{98+112}{2}\right) = \Pr\left\{\text{Bin}\left(122, \frac{1}{2}\right) \geqslant 110\right\} \simeq 0$$

- Sensitivity bound:

$$\begin{aligned}
&\Pr\left\{\text{Bin}\left(122, \frac{1}{1+\Gamma}\right) \geqslant 110\right\} \\
&\leqslant \Pr\left(\frac{T+122}{2} \geqslant 110\right) \\
&\leqslant \Pr\left\{\text{Bin}\left(122, \frac{\Gamma}{1+\Gamma}\right) \geqslant 110\right\}
\end{aligned}$$

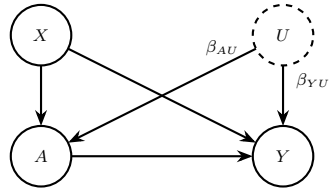
Γ	$p_s = \Gamma/(1+\Gamma)$	LB	UB
1.0	0.5000	0.0000	0.0000
3.0	0.7500	0.0000	0.0000
5.0	0.8333	0.0000	0.0232
5.5	0.8462	0.0000	0.05197
6.0	0.8571	0.0000	0.0969

- Even if $p_s \neq 0.5$ and one pair is more likely to smoke because of U, this unobserved variable must have a $\Gamma > 5.5$-fold difference away from 50/50 randomization to not reject the null hypothesis at $\alpha = 0.05$ that smoking does not cause lung cancer

- In other words, an unmeasured variable must change p_s outside of the range $0.15 \leqslant p_s \leqslant 0.85$ for us to retain the null hypothesis

Amplification of Γ

- Rosenbaum and Silber (2009); assume binary U



- We can re-interpret Γ using the following equation:

$$\Gamma = \frac{\beta_{AU}\beta_{YU} + 1}{\beta_{AU} + \beta_{YU}}$$

where

$$\text{logit}\,\Pr(A = 1|X, U) = \beta_{A0} + \beta_{AX}X + \beta_{AU}U$$

$$\text{logit}\,\Pr(Y = 1|A, X, U) = \beta_{Y0} + \beta_{YA}A + \beta_{YX}X + \beta_{YU}U$$

- The amplification will map a value of Γ into a set or curve of values of (β_{AU}, β_{YU}) such that the one-dimensional analysis for Γ is correct for the entire curve of values of (β_{AU}, β_{YU})

- Therefore, an investigator may perform and display the one-dimensional analysis in terms of Γ, yet interpret that analysis in terms of the two-dimensional (β_{AU}, β_{YU})

- If we set $\mathcal{E} = \beta_{AU} = \beta_{YU}$ (aka evidence value), we have $\mathcal{E} = \Gamma \pm (\Gamma^2 - 1)^{0.5}$

Amplification of Γ

- In the previous smoking example, H_0 is rejected at $\Gamma = 5$
- Even if U increased the odds of smoking by as much as $9.90 = 5 + \sqrt{24}$ times, and simultaneously increased the odds of lung cancer by 9.9 times, we would still have evidence that smoking causes lung cancer
- Even with such a very strong U, we would still be able to conclude that smoking caused lung cancer
- This observational study's conclusion is very **insensitive** to U

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Assignment

- Analyze the dataset `causaldata::close_college`

  While the dataset contains additional variables, we will focus only on the following:
  - $X = (\texttt{black, south, married, smsa, nearc4})$
  - $A = \mathbb{1}(\texttt{educ} \geqslant 16)$ (i.e., whether the individual a 4-year college degree)
  - $Y = \mathbb{1}(\texttt{lwage} > 6.286928)$ (i.e., whether the income is above median)

  We aim to study the effect of having more than a high school degree on the (log) wage

- For detailed variable descriptions/contexts, see `?causaldata::close_college`

- Matching methods to apply:
  - Exact matching; Nearest neighbor matching; Caliper matching; Full matching
  - Perform a sensitivity analysis on the results of one selected pair-matching method

- You may also try 1:M matching, with or without repetition, to achieve better covariate balance

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

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Matching estimators in Observational studies

- Abadie and Imbens (2006) studied the matching estimator for the ATE in observational studies
  - asymptotic properties
  - large-sample confidence intervals

- **Notation:**
  - We observe $\{X_i, A_i, Y_i(1), Y_i(0)\}_{i=1}^{n} \overset{\text{i.i.d.}}{\sim} \{X, A, Y(1), Y(0)\}$
  - X_i represents covariates, A_i is the treatment indicator
  - $Y_i(1)$ and $Y_i(0)$ are the potential outcomes

Point Estimation via Matching

- **One-to-M Matching with Replacement:**
  - For a treated unit i, impute the control potential outcome:

$$Y_i(1) = Y_i, \quad \widehat{Y}_i(0) = \frac{1}{M} \sum_{k \in J_i} Y_k,$$

    where J_i contains indices of the M closest control units based on $d(X_i, X_k)$

  - Similarly, for a control unit i, impute the treated potential outcome:

$$Y_i(0) = Y_i, \quad \widehat{Y}_i(1) = \frac{1}{M} \sum_{k \in J_i} Y_k$$

  - The matching estimator for the ATE is:

$$\widehat{\tau}^m = \frac{1}{n} \sum_{i=1}^{n} \left(\widehat{Y}_i(1) - \widehat{Y}_i(0) \right)$$

Bias in Matching Estimators

- Matching does not guarantee perfect balance in covariates
- So, $\hat{\tau}^m$ has non-negligible bias, especially when X is multidimensional
- Abadie and Imbens (2006) proposed a bias correction term $\widehat{B}$ to adjust for the bias
  - this can be seen as a form of covariate adjustment

Bias Estimation

- Compute the estimated bias: $\widehat{B} = n^{-1} \sum_{i=1}^{n} \widehat{B}_i$ where

$$\widehat{B}_i = (2A_i - 1) \frac{1}{M} \sum_{k \in J_i} \left(\widehat{\mu}_{1-A_i}(X_i) - \widehat{\mu}_{1-A_i}(X_k) \right)$$

- Here, $\widehat{\mu}_1(X)$ and $\widehat{\mu}_0(X)$ are predicted outcomes using regression models (e.g., OLS)

- For a treated unit with $A_i = 1$, the estimated bias is

$$\widehat{B}_i = M^{-1} \sum_{k \in J_i} \{ \widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_k) \}$$

  which corrects the discrepancy in predicted control potential outcomes due to the mismatch in covariates

Bias-Corrected Matching Estimator

- The bias-corrected matching estimator is:

$$\widehat{\tau}^{\text{mbc}} = \widehat{\tau}^{m} - \widehat{B},$$

which has the following linear expansion:

$$\widehat{\tau}^{\text{mbc}} = \frac{1}{n}\sum_{i=1}^{n} \widehat{\psi}_i,$$

where

$$\widehat{\psi}_i = \widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i) + (2A_i - 1)(1 + K_i/M)\{Y_i - \widehat{\mu}_{A_i}(X_i)\}$$

- Here, K_i represents the number of times unit i was used as a match

Variance Estimation

- Motivated by the linear expansion, the variance estimator is

$$\widehat{V}^{\text{mbc}} = \frac{1}{n^2} \sum_{i=1}^{n} \left(\widehat{\psi}_i - \widehat{\tau}^{\text{mbc}} \right)^2$$

  by viewing $\widehat{\tau}^{\text{mbc}}$ as sample averages of the $\widehat{\psi}_i$'s

- This estimator is easy to compute and avoids complex resampling techniques.

- Abadie and Imbens (2006) showed that naive bootstrap resampling does not work for matching estimators

- Otsu and Rai (2017) proposed bootstrapping the $\widehat{\psi}_i$ terms in the linear expansion

- Their approach effectively leads to the variance estimator $\widehat{V}^{\text{mbc}}$, making implementation straightforward

Doubly-Robust estimators

- Define the residual $\widehat{R}_i$ as:

$$\widehat{R}_i = \begin{cases} Y_i - \widehat{\mu}_1(X_i) & \text{if } A_i = 1 \\ Y_i - \widehat{\mu}_0(X_i) & \text{if } A_i = 0 \end{cases}$$

- The standard outcome regression estimator for the average causal effect is:

$$\widehat{\tau}^{\text{reg}} = \frac{1}{n} \sum_{i=1}^{n} \{\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i)\}$$

- The doubly robust estimator combines outcome regression with inverse probability weighting:

$$\widehat{\tau}^{\text{dr}} = \widehat{\tau}^{\text{reg}} + \frac{1}{n} \sum_{i=1}^{n} \left\{ \frac{A_i \widehat{R}_i}{\widehat{e}(X_i)} - \frac{(1 - A_i)\widehat{R}_i}{1 - \widehat{e}(X_i)} \right\}$$

- We will revisit this estimator later

Bias-Corrected Matching Estimator

- Recall that

$$\widehat{\tau}^{\text{mbc}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\widehat{\mu}_1(X_i) - \widehat{\mu}_0(X_i) + (2A_i - 1)\left(1 + \frac{K_i}{M}\right)\{Y_i - \widehat{\mu}_{A_i}(X_i)\right\}$$

- The bias-corrected matching estimator can be reformatted in a way similar to the doubly robust estimator:

$$\widehat{\tau}^{\text{mbc}} = \widehat{\tau}^{\text{reg}} + \frac{1}{n}\sum_{i=1}^{n}\left\{\left(1 + \frac{K_i}{M}\right)A_i\widehat{R}_i - \left(1 + \frac{K_i}{M}\right)(1 - A_i)\widehat{R}_i\right\}$$

- Matching can be viewed as a nonparametric method for estimating the propensity score, and the resulting bias-corrected matching estimator can be viewed as a doubly robust estimator

Matching as a Nonparametric Propensity Score Estimator

- The weight $1 + K_i/M$ should be close to $1/\hat{e}(X_i)$
- If a treated unit has a small $e(X_i)$, it gets matched to many control units, increasing K_i and thus $1 + K_i/M$
- With a fixed M, $1 + K_i/M$ as an estimator for $1/e(X_i)$ will be very noisy
- Allowing M to grow with sample size improves estimation stability
- Lin et al. (2023) formally showed that bias-corrected matching achieves properties similar to doubly robust estimation when M increases appropriately

Matching Estimator for ATT

- **Goal:** Estimate the ATT using matching

- ATT measures the causal effect of treatment among treated individuals:

$$\tau_{\mathsf{T}} = E(Y \mid A = 1) - E\{Y(0) \mid A = 1\}$$

- Matching is used to impute the missing control potential outcomes for treated units, resulting the estimator

$$\widehat{\tau}_{\mathsf{T}}^{\mathsf{m}} = n_1^{-1} \sum_{i=1}^{n} A_i \{Y_i - \widehat{Y}_i(0)\},$$

where $\widehat{Y}_i(0)$ is the imputed control outcome for treated units, computed as:

$$\widehat{Y}_i(0) = \frac{1}{M} \sum_{k \in J_i} Y_k$$

Bias-Corrected ATT Estimator

- $\widehat{\tau}_{\mathsf{T}}^{\mathsf{m}}$ is biased with multidimensional covariates, due to imperfect covariate balance within matched pairs

- Otsu and Rai (2017) proposed a bias correction term:

$$\widehat{B}_{\mathsf{T}} = n_1^{-1} \sum_{i=1}^{n} A_i \widehat{B}_{\mathsf{T},i} \ , \quad \widehat{B}_{\mathsf{T},i} = \frac{1}{M} \sum_{k \in J_i} \{\widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_k)\}$$

- The bias-corrected matching estimator for ATT is $\widehat{\tau}_{\mathsf{T}}^{\mathsf{mbc}} = \widehat{\tau}_{\mathsf{T}}^{\mathsf{m}} - \widehat{B}_{\mathsf{T}}$

$$\widehat{\tau}_{\mathsf{T}}^{\mathsf{mbc}} = \frac{1}{n_1} \sum_{i=1}^{n} A_i \left(Y_i - \frac{1}{M} \sum_{k \in J_i} Y_k \right) - \frac{1}{n_1} \sum_{i=1}^{n} A_i \frac{1}{M} \sum_{k \in J_i} \{\widehat{\mu}_0(X_i) - \widehat{\mu}_0(X_k)\}$$

- It has a linear expansion: $\widehat{\tau}_{\mathsf{T}}^{\mathsf{mbc}} = n_1^{-1} \sum_{i=1}^{n} \widehat{\psi}_{\mathsf{T},i}$, where:

$$\widehat{\psi}_{\mathsf{T},i} = A_i \{Y_i - \widehat{\mu}_0(X_i)\} - (1 - A_i) \frac{K_i}{M} \{Y_i - \widehat{\mu}_0(X_i)\}$$

- By viewing $\widehat{\tau}_{\mathsf{T}}^{\mathsf{mbc}}$ as n/n_1 multiplied by the sample average of the $\widehat{\psi}_{\mathsf{T},i}$'s, the variance estimator is:

$$\widehat{V}_{\mathsf{T}}^{\mathsf{mbc}} = \frac{1}{n_1^2} \sum_{i=1}^{n} \left(\widehat{\psi}_{\mathsf{T},i} - \widehat{\tau}_{\mathsf{T}}^{\mathsf{mbc}} \frac{n_1}{n} \right)^2$$

Comparison with Doubly Robust Estimators

- The outcome regression estimator for ATT is:

$$\widehat{\tau}_{\mathsf{T}}^{\mathsf{reg}} = \frac{1}{n_1} \sum_{i=1}^{n} A_i \{Y_i - \widehat{\mu}_0(X_i)\}$$

- The doubly robust estimator is:

$$\widehat{\tau}_{\mathsf{T}}^{\mathsf{dr}} = \widehat{\tau}_{\mathsf{T}}^{\mathsf{reg}} - \frac{1}{n_1} \sum_{i=1}^{n} \frac{\widehat{e}(X_i)}{1 - \widehat{e}(X_i)} (1 - A_i) \widehat{R}_i$$

- The bias-corrected matching estimator can be rewritten as:

$$\widehat{\tau}_{\mathsf{T}}^{\mathsf{mbc}} = \widehat{\tau}_{\mathsf{T}}^{\mathsf{reg}} - \frac{1}{n_1} \sum_{i=1}^{n} \frac{K_i}{M} (1 - A_i) \widehat{R}_i$$

- Matching essentially uses K_i/M to estimate the odds of the treatment given covariates

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